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TASK	PYTHON BASICS REPORT

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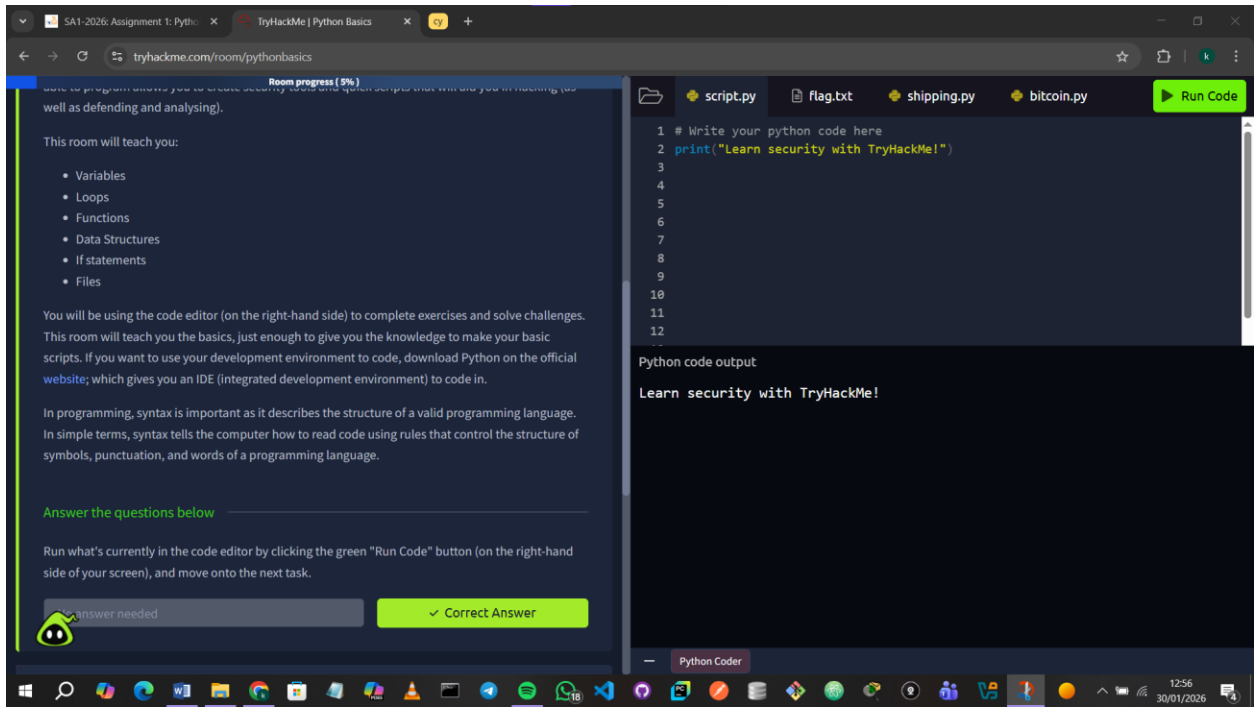
PYTHON BASICS REPORT

Introduction

In this module, I worked hands-on with Python and strengthened my scripting skills for cybersecurity tasks. I explored variables, data types, loops, functions, conditionals, file handling, and library imports, gaining practical experience in automating tasks and developing small tools. This module reinforced my existing Python proficiency while giving me new insights into applying it in a security context.

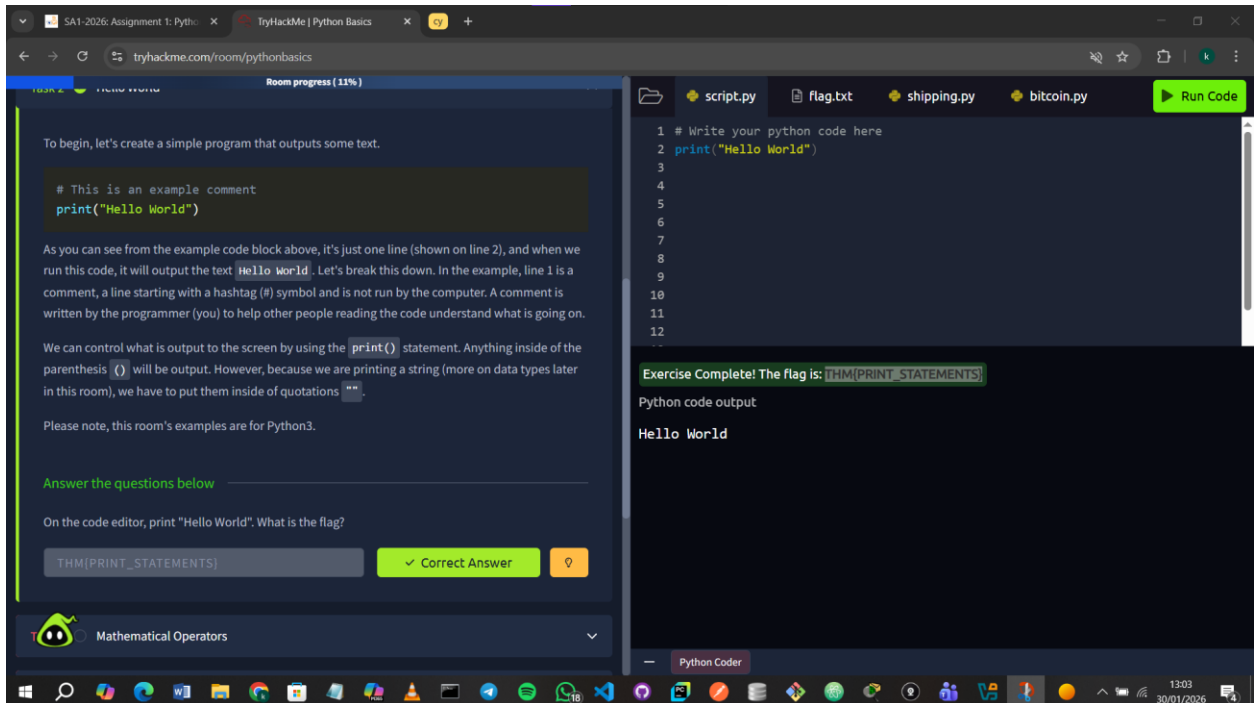
1.0 Introduction to Python

In this section, I learnt the fundamentals of Python and how it can be used for scripting in cybersecurity.



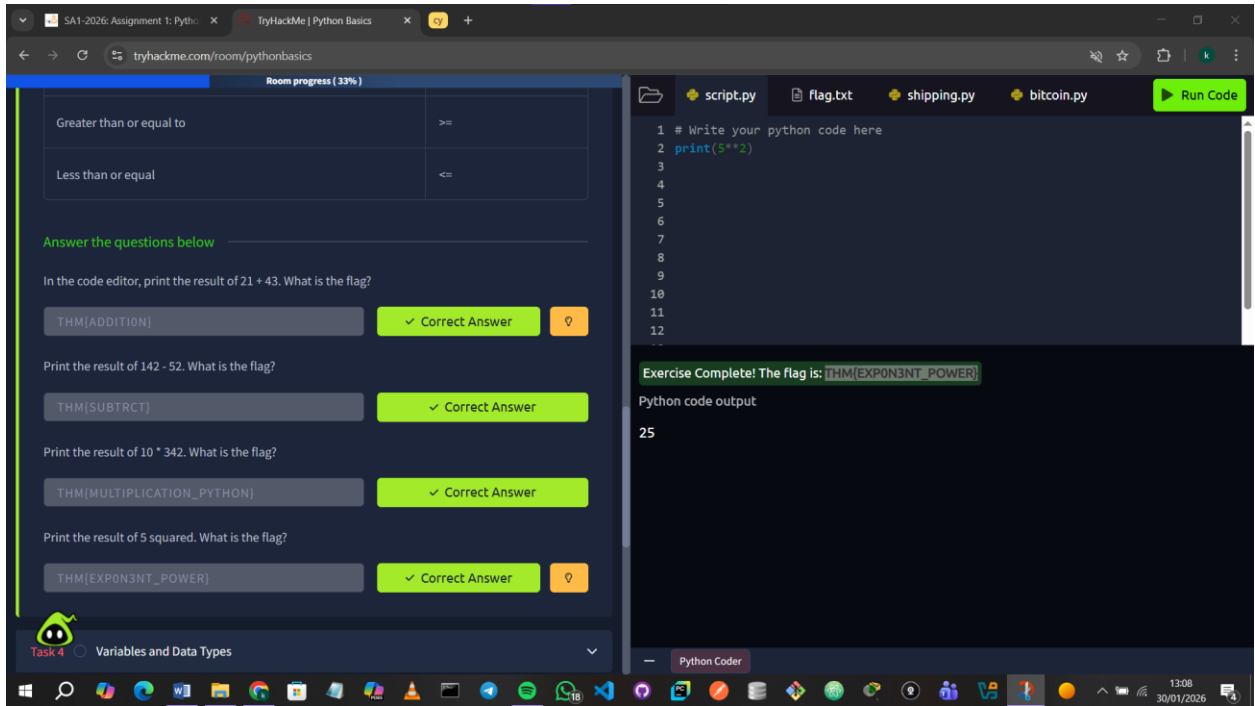
2.0 Hello World

In this section, I learnt how to write a simple Python program using the `print()` function to display output.



3.0 Mathematical operations

In this section, I learnt how Python uses mathematical operators for calculations like addition, subtraction, multiplication, and division. I also learnt about comparison operators, which are used to compare values and are important for making decisions in programs.



4.0 Variables and Data Types

In this section, I learnt how variables are used to store and update data in a program. I also learnt about different data types in Python, such as strings, numbers, Booleans, and lists, and how they define the kind of data a variable can hold.

The screenshot displays the TryHackMe Python Basics room interface. On the left, a table lists movies and their ratings, and a list of tasks is visible. The right panel shows a code editor with Python code and a terminal output.

Movie	Rating	Score	Is it a Star Wars movie?	Characters
Star Wars	9.8	13	True	Alice, Bob
Matrix	8.5	23	False	Charlie
Indiana Jones	6.1	3	False	Daniel, Evie

Answer the questions below

In the code editor, create a variable called height and set its initial value to 200.

No answer needed

On a new line, add 50 to the height variable.

No answer needed

On another new line, print out the value of height. What is the flag that appears?

THM[VARIABLE3]

Task 5 Logical and Boolean Operators

Task 7 Loops

Shipping Project Introduction to If Statements

Python code editor:

```
1 # Write your python code here
2 height = 200
3 height += 50
4 print(height)
5
6
7
8
9
10
11
12
```

137/472 Points earned! Levelling up in progress...

144/472 Points Earned! Levelling up in progress...

Exercise Complete! The flag is: THM[VARIABLE3]

Python code output

250

5.0 Logical and Boolean Operations

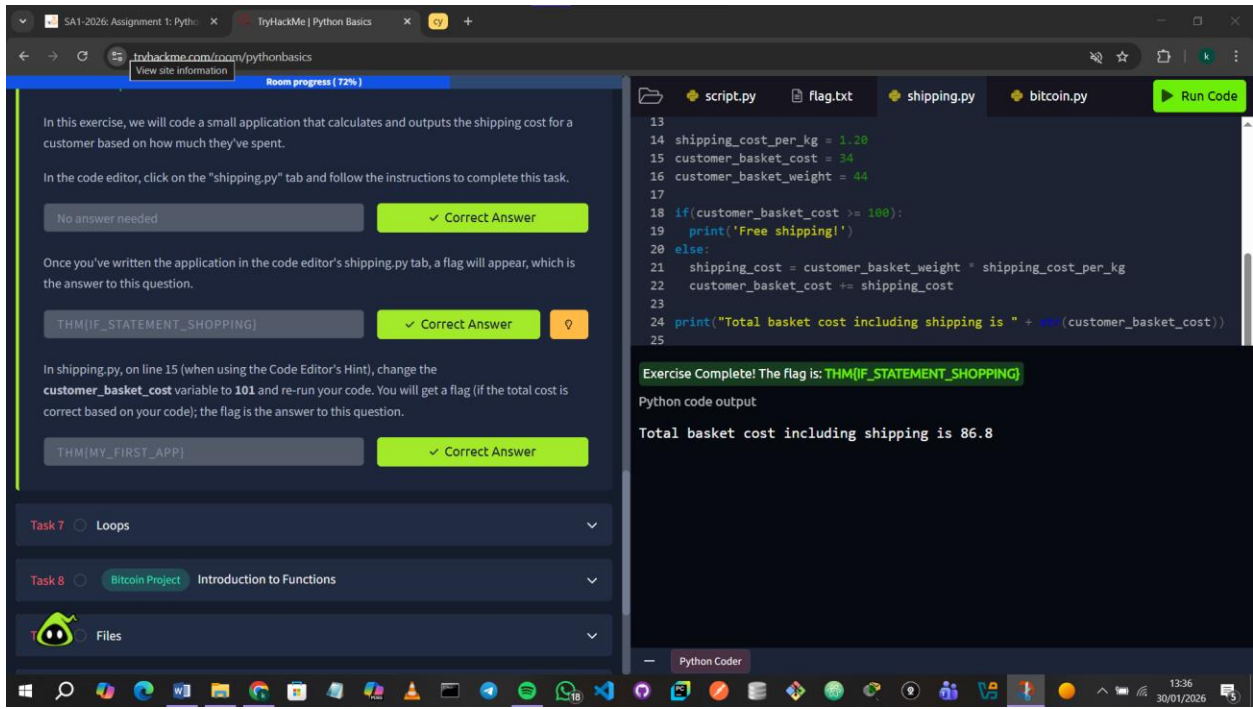
In this section, I learnt how logical and Boolean operators are used to compare values and test conditions in Python. I also learnt how operators like **AND**, **OR**, and **NOT** help combine conditions and control how a program makes decisions based on true or false results.

6.0 Introduction to If Statements

In this section, I learnt how **if statements** allow a program to make decisions based on conditions. I learnt how different blocks of code run depending on whether a condition is true or false, and how indentation and colons define what code belongs inside the if or else statements.

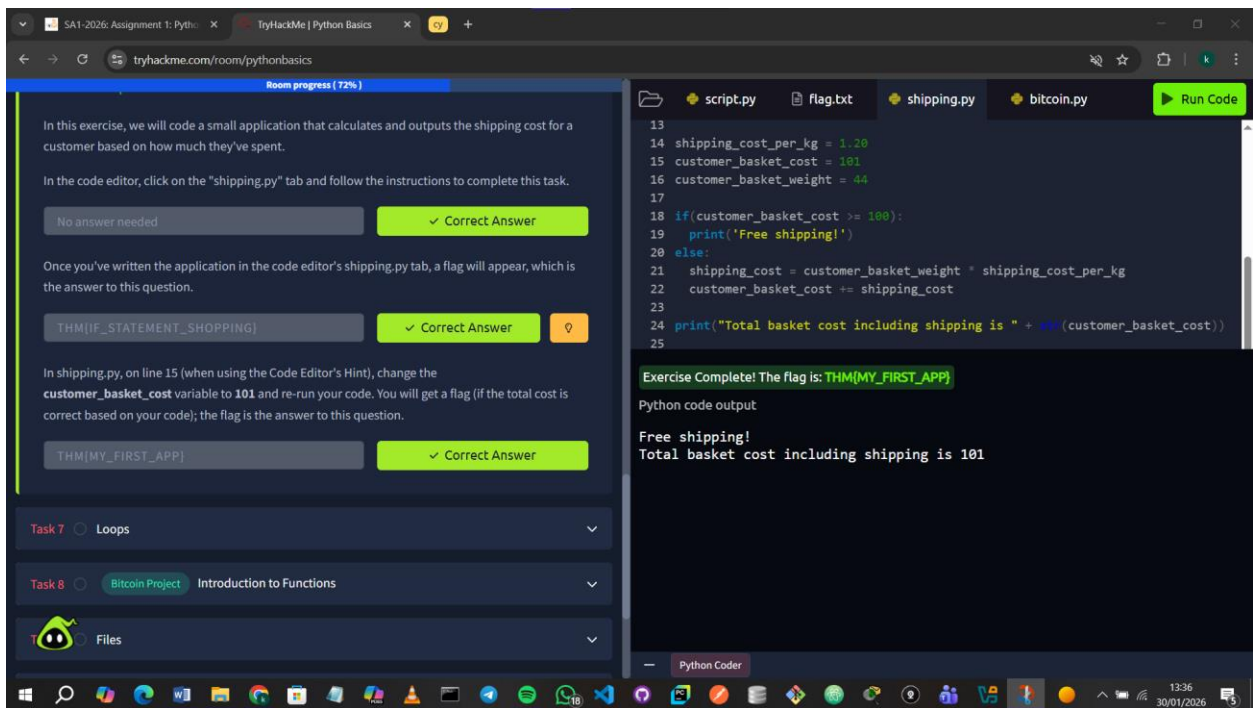
Question: Once you've written the application in the code editor's shipping.py tab, a flag will appear, which is the answer to this question.

Answer: I used if statements to decide whether or not the customer had to pay for shipping.



Question: In shipping.py, on line 15 (when using the Code Editor's Hint), change the customer basket cost variable to 101 and re-run your code. You will get a flag (if the total cost is correct based on your code); the flag is the answer to this question.

Answer:

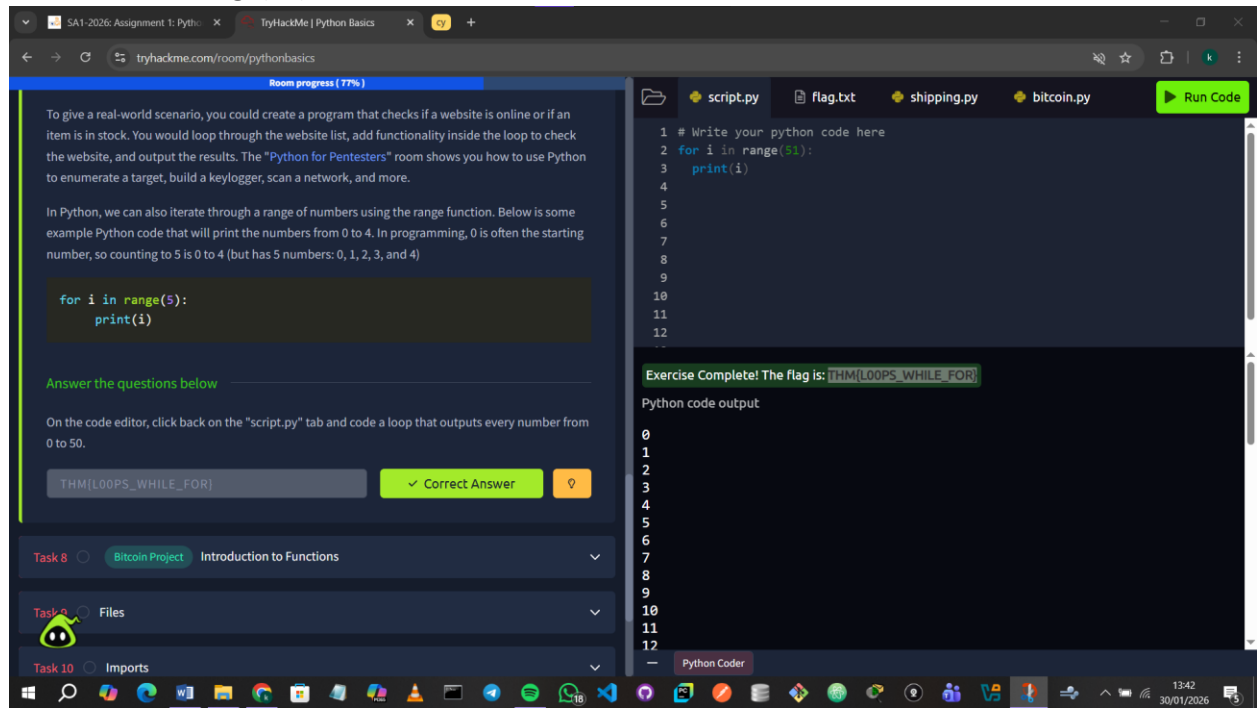


7.0 [Loops](#)

In this section, I learnt how **loops** let a program repeat actions automatically. I learnt how **while loops** run based on a condition and how **for loops** iterate through lists or ranges, which is useful for automating tasks like checking multiple items or targets.

Question: code a loop that outputs every number from 0 to 50.

Answer: I used **range(51)**

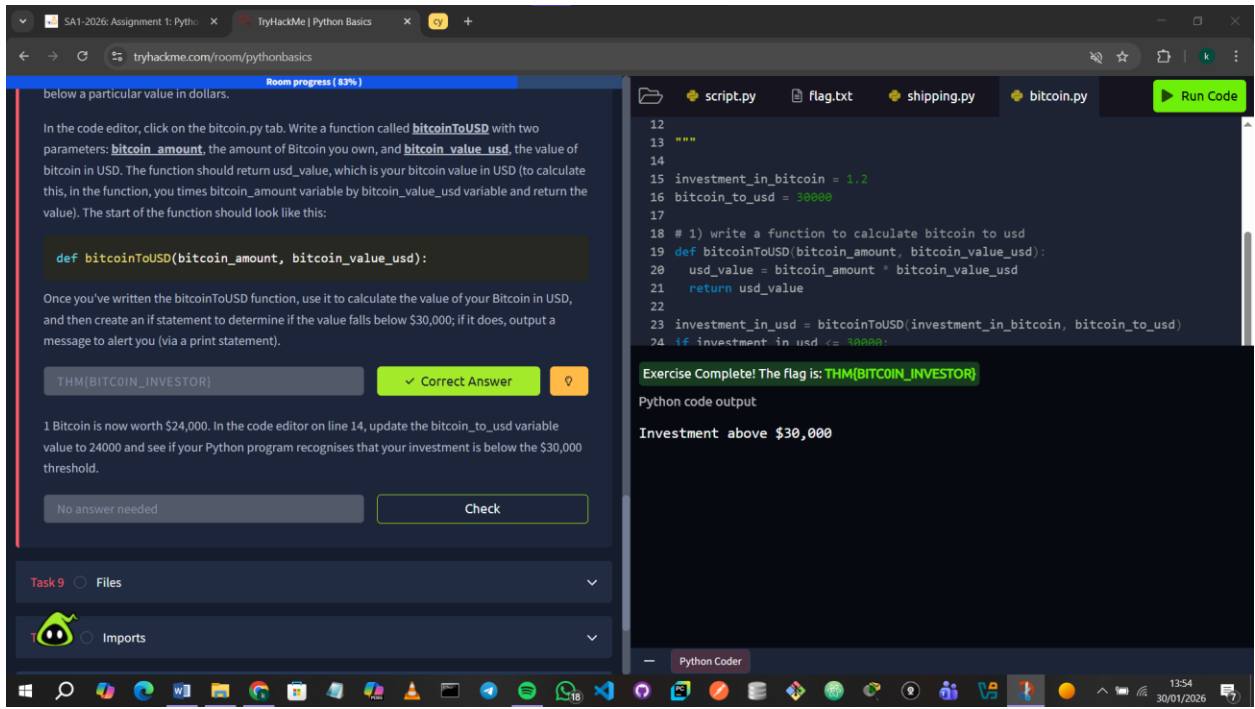


8.0 [Introduction to Functions](#)

In this section, I learnt that **functions** are reusable blocks of code that help avoid repetition.

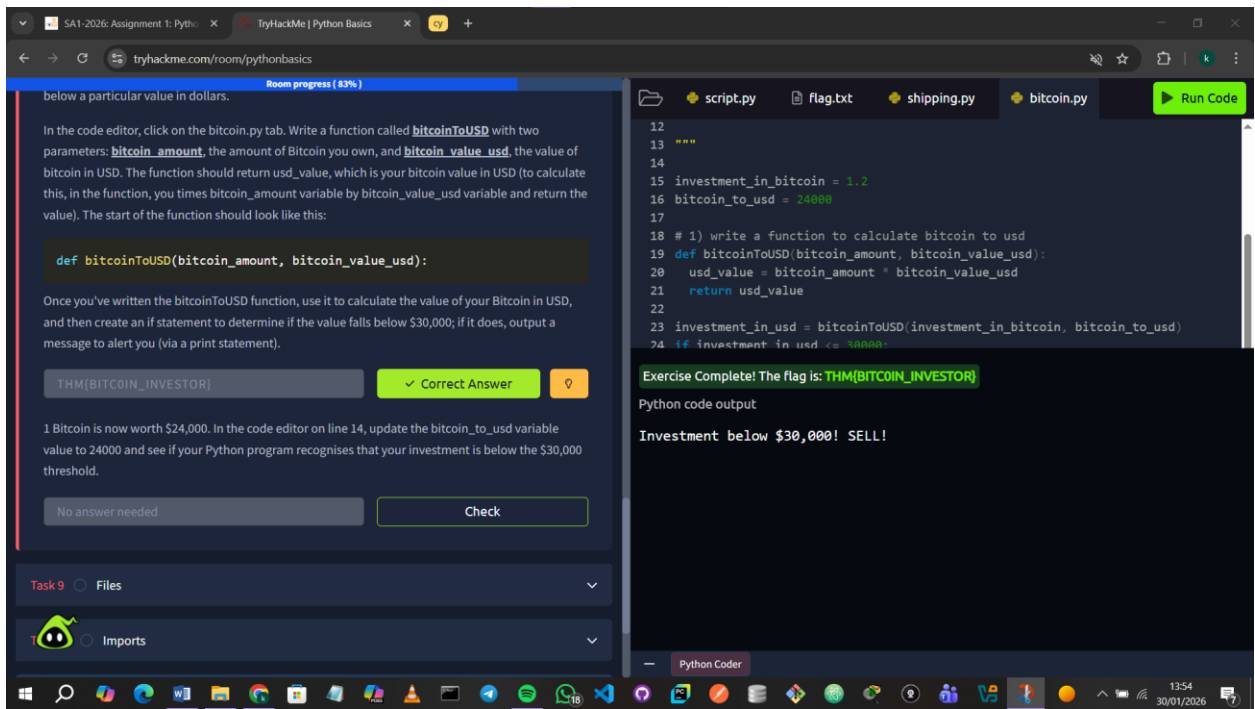
Question: Once you've written the bitcoinToUSD function, use it to calculate the value of your Bitcoin in USD, and then create an if statement to determine if the value falls below \$30,000; if it does, output a message to alert you (via a print statement).

Answer: I made the function then applied if statements to decide what should be printed out.



Question: 1 Bitcoin is now worth \$24,000. In the code editor on line 14, update the bitcoin to usd variable value to 24000 and see if your Python program recognises that your investment is below the \$30,000 threshold.

Answer:

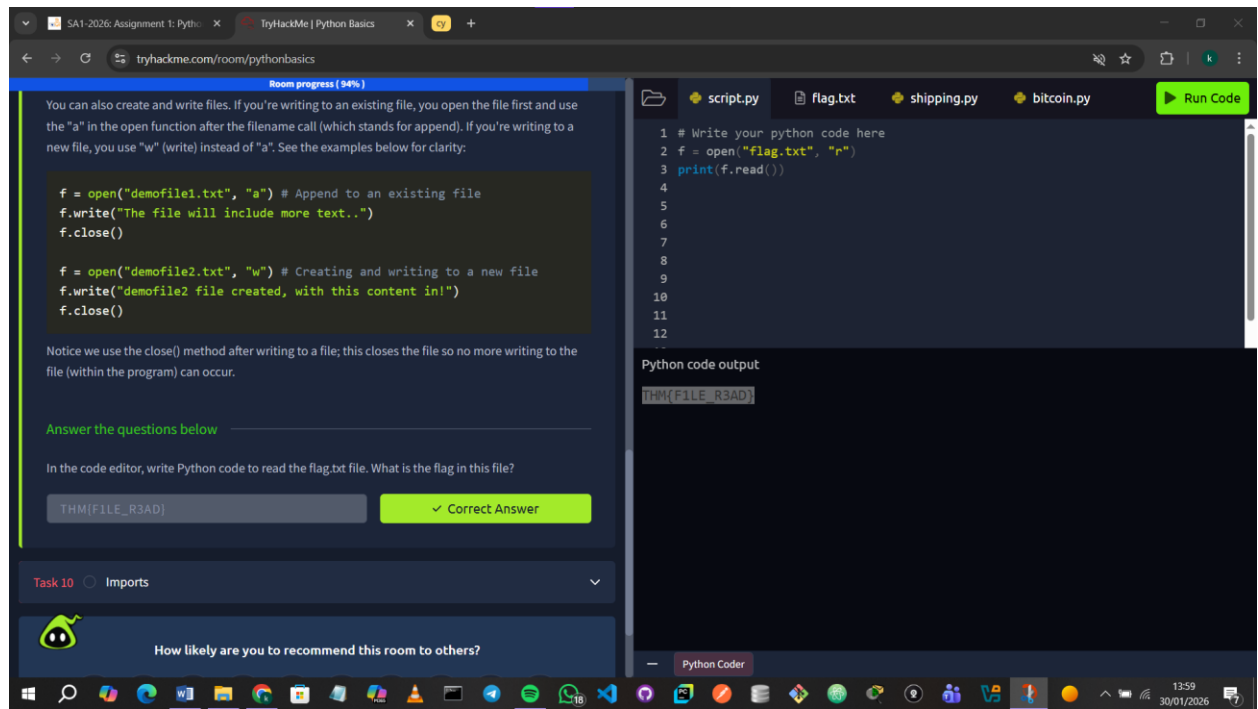


9.0 Files

In this section, I learnt how to **read from and write to files in Python**. I learnt how to open files in different modes (r, a, w), read their contents, write or append data, and properly close files to save changes.

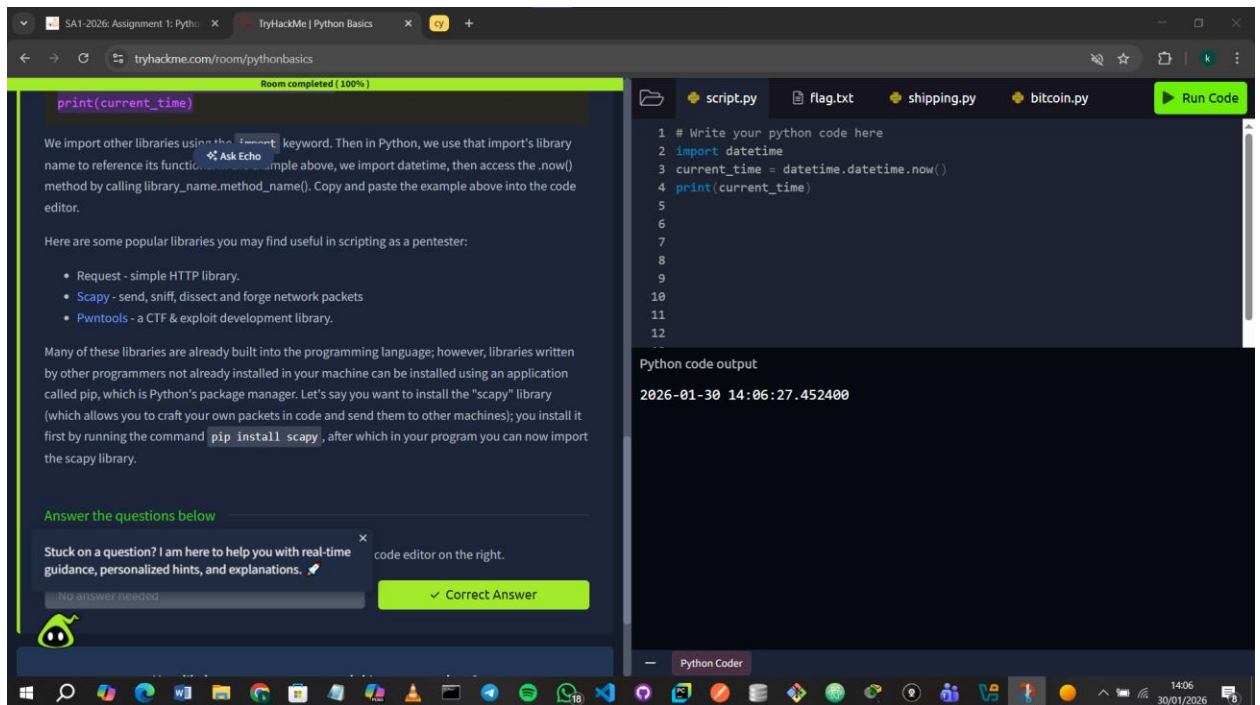
Question: In the code editor, write Python code to read the flag.txt file. What is the flag in this file?

Answer: I used **open** to open the file and **r** to read it then printed it out.



10.0 Imports

In this section, I learnt how to **import and use Python libraries**.



The screenshot shows the TryHackMe Python Basics room interface. On the left, there is a text area with instructions on how to use the `import` keyword and a list of popular libraries: `Request`, `Scapy`, and `Pwntools`. Below this, there is a section titled "Answer the questions below" with a "Correct Answer" button. On the right, there is a code editor with a file explorer showing files like `script.py`, `flag.txt`, `shipping.py`, and `bitcoin.py`. The code editor contains a Python script that imports the `datetime` module and prints the current time. The output of the code is displayed below the editor.

```
print(current_time)
```

We import other libraries using the `import` keyword. Then in Python, we use that import's library name to reference its function. For example, above, we import `datetime`, then access the `.now()` method by calling `library_name.method_name()`. Copy and paste the example above into the code editor.

Here are some popular libraries you may find useful in scripting as a pentester:

- `Request` - simple HTTP library.
- `Scapy` - send, sniff, dissect and forge network packets
- `Pwntools` - a CTF & exploit development library.

Many of these libraries are already built into the programming language; however, libraries written by other programmers not already installed in your machine can be installed using an application called `pip`, which is Python's package manager. Let's say you want to install the "scapy" library (which allows you to craft your own packets in code and send them to other machines); you install it first by running the command `pip install scapy`, after which in your program you can now import the scapy library.

Answer the questions below

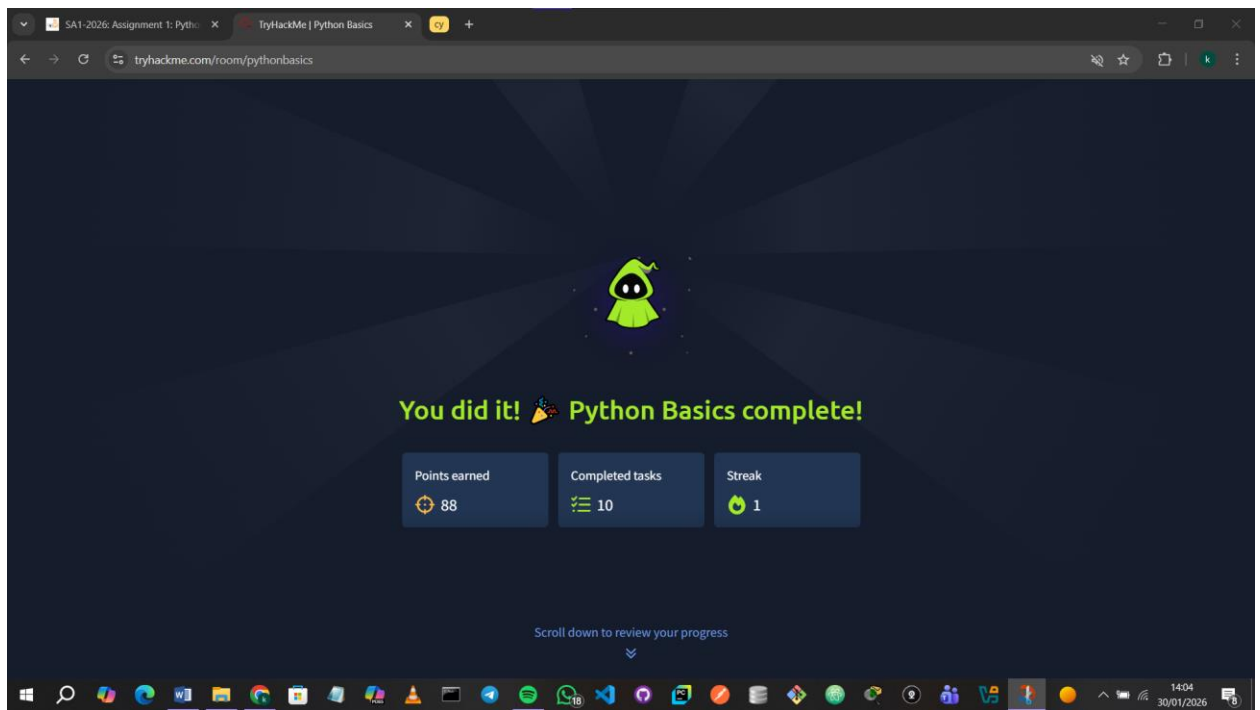
Stuck on a question? I am here to help you with real-time guidance, personalized hints, and explanations.  code editor on the right.

No answer needed ✓ Correct Answer

Python code output

```
1 # Write your python code here
2 import datetime
3 current_time = datetime.datetime.now()
4 print(current_time)
5
6
7
8
9
10
11
12
```

2026-01-30 14:06:27.452400



The screenshot shows the completion screen for the TryHackMe Python Basics room. It features a green robot icon and a message saying "You did it! 🎉 Python Basics complete!". Below this, there are three boxes showing statistics: "Points earned" (88), "Completed tasks" (10), and "Streak" (1). At the bottom, there is a link to "Scroll down to review your progress".

You did it! 🎉 Python Basics complete!

Points earned	Completed tasks	Streak
88	10	1

Scroll down to review your progress

Conclusion

This module covered the essentials of Python, from variables and loops to functions, file handling, and libraries. As someone already proficient in Python, it was a great refresher and helped reinforce best practices for scripting and automation.

